

## C.S. Intro task.

### - Power:

VDD = 5V. The MCP is two chips in one package.

CAN controller + CAN transceiver.

→ VDDA powers the transceiver side.

→ CAN needs to drive CANH to  $\sim 3.5V$  and CANL to  $\sim 1.5V$  from a 2.5V rest point, and you can't produce that from a 3.3V rail.

→ VDDA must be 5V.

### - STM 32:

Runs 2V-3.6V → can't be powered by 5V → Needs a second rail.

### - Decoupling:

\* 0.1µF capacitors on every supply pin

→ 10µF capacitor

- Because chips pull current in small bursts some times.

- Caps sit at the pins as a local reservoir → rails don't flip.



### X7R ceramic

↳ - Need low ESR + ESL, not just the right capacitance

- Ceramic has both.
- Electrolytic would be useless at these speeds.
- Non-polarised → No orientation to make up.

### CAN side:

- TXD → Tx CAN
- RXD → Rx CAN
- Both in the MCP but datasheet says to wire externally

### PESD CAN dual I/Os

- CANH and CANL leave the board over cable.
- Pick up ESD
- Placed at connector → surges stop before UI.

### LDO

↳ - Total load is roughly 25mA

Dissipation:  $(5V - 3.5V) \cdot 0.025 \approx 42mW$

→ which any SOT handles without heating up.

↳ AP2127:

300mA

Everything connected to one GND.

# MCP25625

### 1.3 Typical Application

Figure 1-2 shows an example of a typical application of the MCP25625. In this example, the microcontroller operates at 3.3V.

$V_{DDA}$  supplies the CAN transceiver and must be connected to 5V.

$V_{DD}$ ,  $V_{IO}$  of the MCP25625 are connected to the  $V_{DD}$  of the microcontroller. The digital supply can range from 2.7V to 5.5V. Therefore, the I/O of the MCP25625 is connected directly to the microcontroller, no level shifters are required.

The T<sub>XD</sub> and R<sub>XD</sub> pins of the CAN transceiver must be externally connected to the TxCAN and RxCAN pins of the CAN controller.

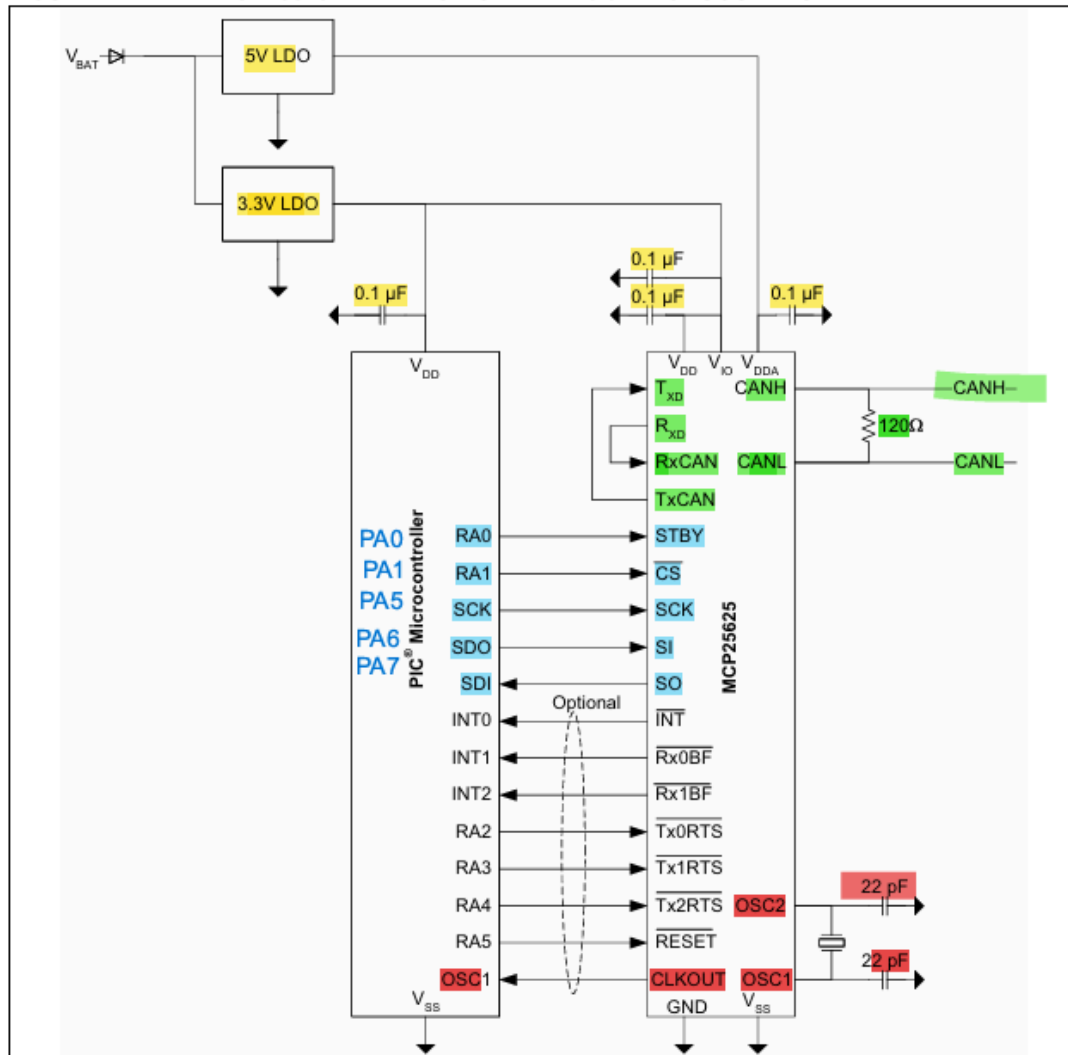
The SPI interface is used to configure and control the CAN controller.

The  $\overline{\text{INT}}$  pin of the MCP25625 signals an interrupt to the microcontroller. Interrupts need to be cleared by the microcontroller through SPI.

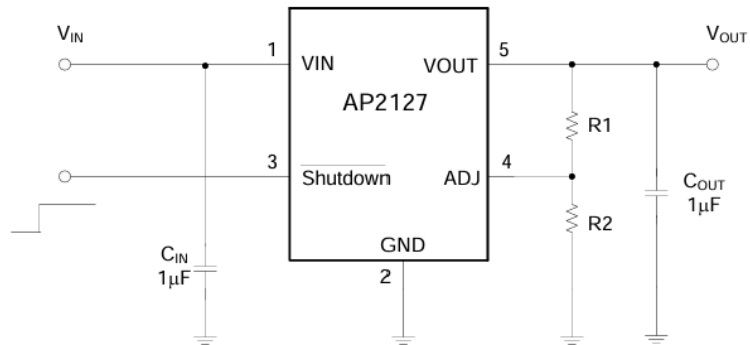
The usage of  $\overline{\text{RxnBF}}$  and  $\overline{\text{TxnRTS}}$  is optional, since the functions of these pins can be accessed through SPI. The RESET pin can optionally be pulled up to the  $V_{DD}$  of the MCP25625 using a 10 k $\Omega$  resistor.

The CLKOUT pin provides the clock to the microcontroller.

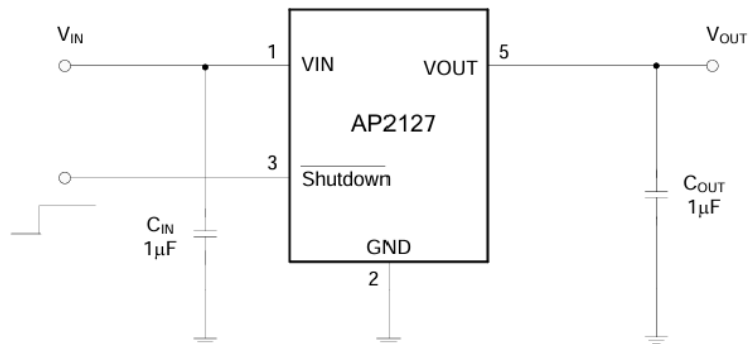
**FIGURE 1-2: MCP25625 INTERFACING WITH A 3.3V MICROCONTROLLER**



# Typical Applications Circuit



$$V_{OUT} = 0.8(1 + R1/R2)V$$

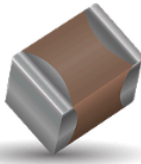


For Fixed Voltage Versions



## X7R Dielectric

### General Specifications



X7R formulations are called "temperature stable" ceramics and fall into EIA Class II materials. X7R is the most popular of these intermediate dielectric constant materials. Its temperature variation of capacitance is within  $\pm 15\%$  from  $-55^\circ\text{C}$  to  $+125^\circ\text{C}$ . This capacitance change is non-linear.

Capacitance for X7R varies under the influence of electrical operating conditions such as voltage and frequency.

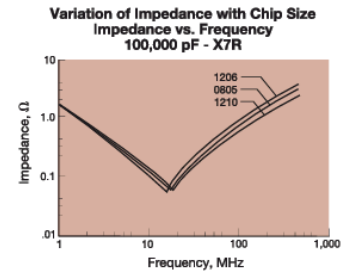
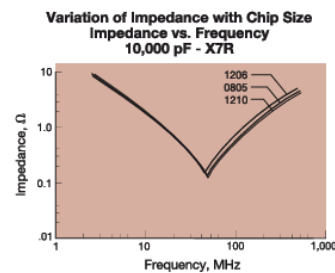
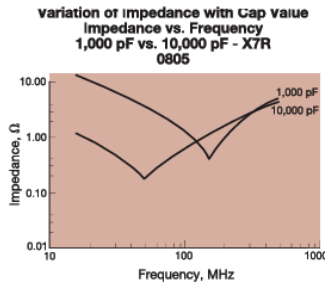
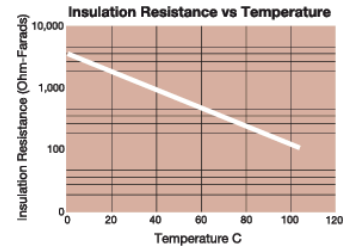
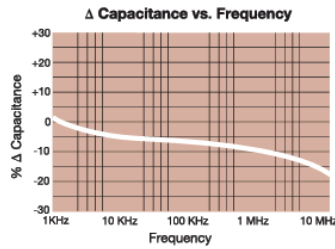
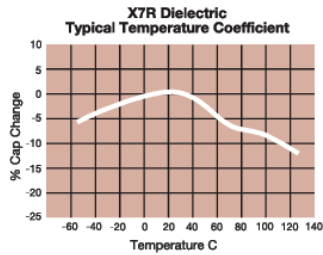
X7R dielectric chip usage covers the broad spectrum of industrial applications where known changes in capacitance due to applied voltages are acceptable.



#### PART NUMBER (SEE PAGE 4 FOR COMPLETE PART NUMBER EXPLANATION)

| 0805              | 5                                                                                                             | C                     | 103                                                               | M                                                                               | A                                        | T                                                                                                                   | 2                                                                                   | A                                      |
|-------------------|---------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------|
| Size<br>(L" x W") | Voltage<br>4V = 4<br>6.3V = 6<br>10V = Z<br>16V = Y<br>25V = 3<br>50V = 5<br>100V = 1<br>200V = 2<br>500V = 7 | Dielectric<br>X7R = C | Capacitance<br>Code (In pF)<br>2 Sig. Digits +<br>Number of Zeros | Capacitance<br>Tolerance<br>J = $\pm 5\%$ *<br>K = $\pm 10\%$<br>M = $\pm 20\%$ | Failure<br>Rate<br>A = Not<br>Applicable | Terminations<br>T = Plated Ni and Sn<br>Z = FLEXITERM**<br>*Optional termination<br>**See FLEXITERM®<br>X7R section | Packaging<br>2 = 7" Reel<br>4 = 13" Reel<br><br>Contact<br>Factory For<br>Multiples | Special<br>Code<br>A = Std.<br>Product |

NOTE: Contact factory for availability of Termination and Tolerance Options for Specific Part Numbers.  
Contact factory for non-specified capacitance values.





# PESD1CAN

## CAN bus ESD protection diode

Rev. 04 — 15 February 2008

Product data sheet

## 1. Product profile

### 1.1 General description

PESD1CAN in a small SOT23 (TO-236AB) Surface-Mounted Device (SMD) plastic package designed to protect two automotive Controller Area Network (CAN) bus lines from the damage caused by ElectroStatic Discharge (ESD) and other transients.

### 1.2 Features

- Due to the integrated diode structure only one small SOT23 package is needed to protect two CAN bus lines
- Max. peak pulse power:  $P_{PP} = 200 \text{ W}$  at  $t_p = 8/20 \mu\text{s}$
- Low clamping voltage:  $V_{CL} = 40 \text{ V}$  at  $I_{PP} = 1 \text{ A}$
- Ultra low leakage current:  $I_{RM} < 1 \text{ nA}$
- Typ. diode capacitance matching:  $\Delta C_d/C_d = 0.1 \%$
- ESD protection up to 23 kV
- IEC 61000-4-2, level 4 (ESD)
- IEC 61000-4-5 (surge);  $I_{PP} = 3 \text{ A}$  at  $t_p = 8/20 \mu\text{s}$
- Small SMD plastic package

### 1.3 Applications

- CAN bus protection
- Automotive applications

### 1.4 Quick reference data

**Table 1. Quick reference data**  
*T<sub>amb</sub> = 25 °C unless otherwise specified.*

| Symbol           | Parameter                | Conditions                      | Min | Typ | Max | Unit |
|------------------|--------------------------|---------------------------------|-----|-----|-----|------|
| <b>Per diode</b> |                          |                                 |     |     |     |      |
| V <sub>RWM</sub> | reverse standoff voltage |                                 | -   | -   | 24  | V    |
| C <sub>d</sub>   | diode capacitance        | f = 1 MHz; V <sub>R</sub> = 0 V | -   | 11  | 17  | pF   |

**Table 12. Pin assignment and description**

| Pin  |         |        |        | Pin name<br>(function upon reset) | Pin type | I/O structure | Note   | Alternate functions                                    | Additional functions              |
|------|---------|--------|--------|-----------------------------------|----------|---------------|--------|--------------------------------------------------------|-----------------------------------|
| SO8N | TSSOP20 | LQFP32 | LQFP48 |                                   |          |               |        |                                                        |                                   |
| -    | -       | -      | 1      | PC13                              | I/O      | FT            | (1)(2) | TIM1_BK                                                | TAMP_IN1, RTC_TS, RTC_OUT1, WKUP2 |
| -    | -       | -      | 2      | PC14-<br>OSC32_IN<br>(PC14)       | I/O      | FT            | (1)(2) | TIM1_BK2                                               | OSC32_IN                          |
| 1    | 2       | 2      | -      | PC14-<br>OSC32_IN<br>(PC14)       | I/O      | FT            | (1)(2) | TIM1_BK2                                               | OSC32_IN, OSC_IN                  |
| -    | 3       | 3      | 3      | PC15-<br>OSC32_OUT<br>(PC15)      | I/O      | FT            | (1)(2) | OSC32_EN, OSC_EN                                       | OSC32_OUT                         |
| -    | -       | -      | 4      | VBAT                              | S        | -             | -      | -                                                      | VBAT                              |
| -    | -       | -      | 5      | VREF+                             | S        | -             | -      | -                                                      | -                                 |
| 2    | 4       | 4      | 6      | VDD/VDDA                          | S        | -             | -      | -                                                      | -                                 |
| 3    | 5       | 5      | 7      | VSS/VSSA                          | S        | -             | -      | -                                                      | -                                 |
| -    | -       | -      | 8      | PF0-OSC_IN<br>(PF0)               | I/O      | FT            | -      | TIM14_CH1                                              | OSC_IN                            |
| -    | -       | -      | 9      | PF1-<br>OSC_OUT<br>(PF1)          | I/O      | FT            | -      | OSC_EN                                                 | OSC_OUT                           |
| 4    | 6       | 6      | 10     | NRST                              | I/O      | RST           | (3)    | -                                                      | NRST                              |
| -    | 7       | 7      | 11     | PA0                               | I/O      | FT_a          | (3)    | SPI2_SCK, USART2_CTS,                                  | ADC_IN0, TAMP_IN2, WKUP1          |
| -    | 8       | 8      | 12     | PA1                               | I/O      | FT_ea         | (3)    | SPI1_SCK/I2S1_CK, USART2_RTS_DE_CK, I2C1_SMB, EVENTOUT | ADC_IN1                           |
| -    | 9       | 9      | 13     | PA2                               | I/O      | FT_a          | (3)    | SPI1_MOSI/I2S1_SD, USART2_TX,                          | ADC_IN2, WKUP4, LSCO              |
| -    | 10      | 10     | 14     | PA3                               | I/O      | FT_ea         | -      | SPI2_MISO, USART2_RX, EVENTOUT                         | ADC_IN3                           |
| -    | -       | -      | 15     | PA4                               | I/O      | FT_a          | -      | SPI1_NSS/I2S1_WS, SPI2_MOSI, TIM14_CH1, EVENTOUT       | ADC_IN4, RTC_OUT2                 |

Table 12. Pin assignment and description (continued)

| Pin  |         |        |        | Pin name<br>(function upon reset) | Pin type | I/O structure | Note | Alternate functions                                                | Additional functions                             |
|------|---------|--------|--------|-----------------------------------|----------|---------------|------|--------------------------------------------------------------------|--------------------------------------------------|
| SO8N | TSSOP20 | LQFP32 | LQFP48 |                                   |          |               |      |                                                                    |                                                  |
| -    | 11      | 11     | -      | PA4                               | I/O      | FT_a          | -    | SPI1_NSS/I2S1_WS,<br>SPI2_MOSI, TIM14_CH1,<br>EVENTOUT             | ADC_IN4,<br>TAMP_IN1, RTC_TS,<br>RTC_OUT1, WKUP2 |
| -    | 12      | 12     | 16     | PA5                               | I/O      | FT_ea         | -    | SPI1_SCK/I2S1_CK,<br>EVENTOUT                                      | ADC_IN5                                          |
| -    | 13      | 13     | 17     | PA6                               | I/O      | FT_ea         | -    | SPI1_MISO/I2S1_MCK,<br>TIM3_CH1, TIM1_BK,<br>TIM16_CH1             | ADC_IN6                                          |
| -    | 14      | 14     | 18     | PA7                               | I/O      | FT_a          | -    | SPI1_MOSI/I2S1_SD,<br>TIM3_CH2, TIM1_CH1N,<br>TIM14_CH1, TIM17_CH1 | ADC_IN7                                          |
| 5    | 15      | 15     | 19     | PB0                               | I/O      | FT_ea         | -    | SPI1_NSS/I2S1_WS,<br>TIM3_CH3, TIM1_CH2N                           | ADC_IN8                                          |
| 5    | 15      | 16     | 20     | PB1                               | I/O      | FT_ea         | -    | TIM14_CH1, TIM3_CH4,<br>TIM1_CH3N, EVENTOUT                        | ADC_IN9                                          |
| -    | 15      | 17     | 21     | PB2                               | I/O      | FT_ea         | -    | SPI2_MISO, EVENTOUT                                                | ADC_IN10                                         |
| -    | -       | -      | 22     | PB10                              | I/O      | FT_fa         | -    | SPI2_SCK, I2C2_SCL                                                 | ADC_IN11                                         |
| -    | -       | -      | 23     | PB11                              | I/O      | FT_fa         | -    | SPI2_MOSI, I2C2_SDA                                                | ADC_IN15                                         |
| -    | -       | -      | 24     | PB12                              | I/O      | FT_a          | -    | SPI2_NSS, TIM1_BK,<br>EVENTOUT                                     | ADC_IN16                                         |
| -    | -       | -      | 25     | PB13                              | I/O      | FT_f          | -    | SPI2_SCK, TIM1_CH1N,<br>I2C2_SCL, EVENTOUT                         | -                                                |
| -    | -       | -      | 26     | PB14                              | I/O      | FT_f          | -    | SPI2_MISO, TIM1_CH2N,<br>I2C2_SDA, EVENTOUT                        | -                                                |
| -    | -       | -      | 27     | PB15                              | I/O      | FT            | -    | SPI2_MOSI, TIM1_CH3N,<br>EVENTOUT                                  | RTC_REFIN                                        |
| 5    | 15      | 18     | 28     | PA8                               | I/O      | FT            | -    | MCO, SPI2_NSS, TIM1_CH1,<br>EVENTOUT                               | -                                                |
| -    | -       | 19     | 29     | PA9                               | I/O      | FT_f          | -    | MCO, USART1_TX,<br>TIM1_CH2, SPI2_MISO,<br>I2C1_SCL, EVENTOUT      | -                                                |
| -    | -       | 20     | 30     | PC6                               | I/O      | FT            | -    | TIM3_CH1                                                           | -                                                |
| -    | -       | -      | 31     | PC7                               | I/O      | FT            | -    | TIM3_CH2                                                           | -                                                |



# MCP25625

## 1.2 Pin Out Description

The descriptions of the pins are listed in [Table 1-1](#).

**TABLE 1-1: MCP25625 PIN DESCRIPTION**

| Pin Name         | 6x6 QFN | SSOP | Block <sup>(1)</sup> | Pin Type | Description                                |
|------------------|---------|------|----------------------|----------|--------------------------------------------|
| V <sub>IO</sub>  | 11      | 1    | CAN Transceiver      | P        | Digital I/O Supply Pin for CAN Transceiver |
| NC               | 14      | 2    | —                    | —        | No Connection                              |
| CANL             | 12      | 3    | CAN Transceiver      | HV I/O   | CAN Low-Level Voltage I/O                  |
| CANH             | 13      | 4    | CAN Transceiver      | HV I/O   | CAN High-Level Voltage I/O                 |
| STBY             | 15      | 5    | CAN Transceiver      | I        | Standby Mode Input                         |
| Tx1RTS           | 8       | 6    | CAN Controller       | I        | TXB1 Request-to-Send                       |
| Tx2RTS           | 9       | 7    | CAN Controller       | I        | TXB2 Request-to-Send                       |
| OSC2             | 20      | 8    | CAN Controller       | O        | External Oscillator Output                 |
| OSC1             | 21      | 9    | CAN Controller       | I        | External Oscillator Input                  |
| GND              | 22      | 10   | CAN Controller       | P        | Ground                                     |
| Rx1BF            | 23      | 11   | CAN Controller       | O        | RxB1 Interrupt                             |
| Rx0BF            | 24      | 12   | CAN Controller       | O        | RxB0 Interrupt                             |
| INT              | 25      | 13   | CAN Controller       | O        | Interrupt Output                           |
| SCK              | 26      | 14   | CAN Controller       | I        | SPI Clock Input                            |
| SI               | 27      | 15   | CAN Controller       | I        | SPI Data Input                             |
| SO               | 28      | 16   | CAN Controller       | O        | SPI Data Output                            |
| CS               | 1       | 17   | CAN Controller       | I        | SPI Chip Select Input                      |
| RESET            | 2       | 18   | CAN Controller       | I        | Reset Input                                |
| V <sub>DD</sub>  | 3       | 19   | CAN Controller       | P        | Power for CAN Controller                   |
| TxCAN            | 4       | 20   | CAN Controller       | O        | Transmit Output to CAN Transceiver         |
| RxCAN            | 5       | 21   | CAN Controller       | I        | Receive Input from CAN Transceiver         |
| CLKOUT           | 6       | 22   | CAN Controller       | O        | Clock Output/SOF                           |
| Tx0RTS           | 7       | 23   | CAN Controller       | I        | TXB0 Request-to-Send                       |
| T <sub>XD</sub>  | 16      | 24   | CAN Transceiver      | I        | Transmit Data Input from CAN Controller    |
| NC               | 17      | 25   | —                    | —        | No Connection                              |
| V <sub>SS</sub>  | 18      | 26   | CAN Transceiver      | P        | Ground                                     |
| V <sub>DDA</sub> | 19      | 27   | CAN Transceiver      | P        | Power for CAN Transceiver                  |
| R <sub>XD</sub>  | 10      | 28   | CAN Transceiver      | O        | Receive Data Output to CAN Controller      |
| EP               | 29      | —    | —                    | —        | Exposed Thermal Pad                        |

**Legend:** P = Power, I = Input, O = Output, HV = High Voltage.

**Note 1:** See [Section 3.0 “CAN Controller”](#) and [Section 6.0 “CAN Transceiver”](#) for further information.

time and stored at a given memory location.

## 3.7 Power supply management

### 3.7.1 Power supply schemes

The STM32G030x6/x8 devices require a 2.0 V to 3.6 V operating supply voltage ( $V_{DD}$ ). Several different power supplies are provided to specific peripherals:

- $V_{DD} = 2.0$  to  $3.6$  V  
 $V_{DD}$  is the external power supply for the internal regulator and the system analog such as reset, power management and internal clocks. It is provided externally through VDD/VDDA pin.
- $V_{DDA} = 2.0$  V to  $3.6$  V  
 $V_{DDA}$  is the analog power supply for the A/D converter.  $V_{DDA}$  voltage level is identical to  $V_{DD}$  voltage as it is provided externally through VDD/VDDA pin.
- $V_{DDIO1} = V_{DD}$   
 $V_{DDIO1}$  is the power supply for the I/Os.  $V_{DDIO1}$  voltage level is identical to  $V_{DD}$  voltage as it is provided externally through VDD/VDDA pin.